VICTORIA HIGH SCHOOL –IGANGA

MATHEMATICS DEPARTMENT

PRESENTS

THE GRAND SEMINAR

UGANDA ADVANCED CERTIFICATE OF EDUCATION (UACE)

PRINCIPAL MATHEMATICS P425/1 & P425/2

DATE: SUNDAY, 27TH SEPTEMBER 2026

P425/1 PRINCIPAL MATHEMATICS PAPER 1 ASSESSMENT FORMAT					
SECTION	PARTS	THEME/ CONSTRUCT	TOPICS	NUMBER OF ITEMS SET	ITEMS ATTEMPTED
A	COMPULSORY	GEOMETRY ITEM 1	1.Straight lines 2.Locus 3.Circle 4.Parabola 5.Ellipse 6.Vectors 7.Trigonometry	1	1
B	I	ALGEBRA ITEM 2 AND ITEM 3	1.Equations and inequalities 2.Numerical concepts 3.Permutations and Combinations 4.Series 5.Complex Numbers	2	1
	II	CALCLUS ITEM 4 AND ITEM 5	1. Partial Fractions 3. Differentiation 4.Integration 5.Differential equations	2	1
TOTAL ITEMS				5	3

P425/1
PRINCIPAL MATHEMATICS
Paper 1 2026

3 hours

UGANDA NATIONAL EXAMINATIONS BOARD
Uganda Advanced Certificate of Education
PRINCIPAL MATHEMATICS
Paper 1
3 hours

INSTRUCTIONS TO CANDIDATES

This examination paper consists of **two** *sections;* **A** *and* **B**. *It has* **five** examination *items***.**

Section **A** *consists of* **one compulsory** *item.*

Section **B** *has two parts,* **I** *and* **II** *each with two optional items. Respond to only* **two** *items from Section* **B**, *choosing* **one** *item from* **each** *of the parts,* **I** *and* **II**.

Respond to **three** *items* **in all**.

Any additional item(s) responded to will **not** *be scored.*

Begin each item on a fresh page.

All *necessary working* **must** *be shown clearly.*

Graph paper is provided.

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

THEME ONE: GEOMETRY (SECTION A)

ITEM: ONE

In a certain school, there has been a growing concern about students encroaching on teachers' privacy in the teachers' quarters, boys and girls mixing at the swimming pool, and unnecessary movements during class time. To address the concerns, the administration has come up with the following plans:

Students move from their dormitories located at A(1, 2, 3) to their classes located at C(2, 4, 3) along a straight pathway. All coordinates are in meters.

The teachers' quarters is a restricted area defined by:

$2x - y + 2z = 24$

The administration wants to construct a perimeter wall for privacy, but this will only be possible if the pathway doesn't pass through the restricted area and the distance from the pathway to the restricted area is at least 3 meters.

The school swimming pool is circular with boundaries at A(1, 2)m, B(7, 4)m, and C(4, −2)m on the school map with all coordinates being in metres. To prevent mixing of boys and girls, two separate straight pathways will be constructed from the washrooms located at R(10, 5)m to touch the pool at two different points, one for girls and the other for boys. The administration has a budget which can construct a total length not exceeding 25 metres.

The administration has also resolved to install a rotational CCTV camera whose acute angle of rotation is modeled by,

$2\ sin2\theta + cos2\theta = 1$

For this camera to monitor all the target area, its acute angle should be in the range $55° < \theta < 65°$

Task:

Help the school administration to make valid decisions using your mathematics skills.

ITEM: TWO

An IT company in Mbale city is designing a drone-navigation and signal-processing system for agricultural monitoring. Two drones were launched and their movement from the control station is represented by vectors. They were launched using different routes and at one point they are supposed to meet.

One of the drones took the route described by the vector equation:

$r_1 = (0, -1, -3) + \mu(1, 3, 6)$

while the other drone took the route described by equation:

$r_2 = (-2, 1, 1) + \tau(1, 1, 2)$

The controller is not sure where the two drones will meet and the angle between the two routes if they are to meet.

The same company was asked to design a rotating spotlight to shine on the students performing during their cultural galas as they move on stage. They discovered that the brightness of the light at any instant depends on the angle of rotation. From the design notes, students observed that when the rotation angle is θ (where θ is an acute angle), the brightness after three rotations can be expressed as $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$. Before presenting their project to the judges, the company was tasked to confirm that this relation is mathematically correct and to determine the value of θ when $3\cos\theta - 4\sin\theta = 5$, but lacked an idea on how to confirm it.

The same company plans to install a modern signal transmission system in a newly developed Town council. The engineers in the company decide to use different mathematical models to design the system efficiently. At the Centre of the Town council, a circular recreational park is constructed such that its boundary is represented by the equation $x^2 + y^2 + 5x - 6y = 5$. Far away from the Town council, an elliptical racing track is constructed for testing autonomous vehicles. The track has equation:

$x^2 + 4y^2 = 100$

Task:

a) Help the controller out of the challenge.

b) As a student of pure mathematics, help the company clearly verify that the two equations are equal and find the value of θ.

c) Help the company determine the Centre and radius of the circular park and determine the eccentricity of the ellipse.

ITEM: THREE

An extension worker mapped a village demonstration farm on the cartesian coordinates in three dimensions. He installed a sprinkler at point S(1, 3, 2) inside a green house. The green house roof is the plane $3x - y + 2z = 7$. A water pipe which runs along the line $r = (2 + \mu)i + (-1 + 2\mu)j + (1 - 2\mu)k$ enters the green house roof at point P. The extension worker wishes to move the sprinkler closer if the distance between S and P is more than 5 m.

Two boundaries of the sprinkler spray zone on the green house floor have equations $y = 2x - 1$ and $3x + 4y = 8$. The extension worker wishes to change the nozzle if the angle between the two spray boundaries is less than 50°.

During calibration, the sprinkler rotates through angle x which satisfies the equation $2\,sinx + 5\,cosx - 1 = 0$; ($0° \leq x \leq 180°$). If the angle is less than 100°, the sprinkler will be adjusted.

A faulty nozzle sprays water such that it hits the floor at a point Q(x, y) whose distance from the point A(2, 1) is twice its distance from the point B(5, 3). The extension worker wishes to relocate a plant at (7, 4) if water can't reach it.

Task:

Based on mathematical computations, advise the extension worker on the decisions to take.

ITEM: FOUR

Amina is a final-year land-surveying student on industrial placement with a firm that has been contracted to design a link road joining three trading centres, P, Q and R, located at (1, 2), (−3, 4) and (5, −2) respectively. A water-tower is to be sited at a point equidistant from all three centres. The client has offered two design options ; a circular access road or a triangular one and wants the shorter of the two built.

Task:

(a) Determine whether a triangular road through P, Q and R is geometrically possible.

(b) Derive the equation of the circle Amina can use to model the circular road.

(c) Recommend, with supporting calculations, which design the firm should build, and state where the water-tower should be sited on that design.

THEME TWO: ALGEBRA (SECTION B PART I)

ITEM: FIVE

An engineering company has been contracted to design a roof for a stadium pavilion. The company's engineers have proposed a design modeled by the quadratic function;

$y = Px^2 + Qx + R$

Where P, Q, R are constants and *x* and *y* represent the coordinates of the roof profile in meters.

For the pavilion layout, the engineers identified key target points A(−1, −4)m, B(1, 8)m, and C(2, 11)m that the roof structure must cover. The management will only approve the final design if the curve passes through the point D(3, 12)m, which is expected to be the highest point (Vertex) of the roof profile.

The engineers also plan to use uniform type of iron sheets to roof the entire pavilion. These sheets are manufactured in two different lengths:

$L_1 = 2^{2n-7}$ *metres and* $L_2 = 3(2^{n-4})$ *metres*

Where L_1 is exactly 2 metres longer than L_2 and n represents the material Strength Index of the sheets. Structurally, these sheets can only withstand wind forces and seasonal heat if their material strength index is more than 3.

Furthermore, these sheets expand when heated. This thermal expansion effect is modeled by the formula:

$E = (1 + 2x)^5$

Where x is the ratio of the temperature directly under the roof to the ambient temperature above the roof during heat waves. On a peak sunny day, the temperature under the roof is 20°C, while the passing heatwave reaches 40°C. The sheets will only be recommended for installation if the calculated heating effect, E evaluated using only the first four terms of its expansion does not exceed 27.

Task:

Analyze mathematically the structural parameter provided and help the engineers make valid decisions.

ITEM: SIX

A farmers' cooperative wants to allocate resources fairly, schedule its harvest festival efficiently, and award prizes transparently.

For fair allocation, 19 times the number of sacks given to each farmer, minus the square of that number, must be at least 84.

The organisers have 6 field helpers whom they wish to reward with at most 3 prizes. The prizes for the 3rd, 2nd and 1st positions follow the terms of the expansion of $(1 + 4x)^6$ in ascending powers of x, where x represents \$1.

The organisers also plan to invite 38 guests, each arriving exactly 6 minutes after the previous one, with the first guest arriving at 7:30 a.m. The chief guest arrives last, and the festival opens 10 minutes after that arrival.

Task:

(a) Find the range of the number of sacks each farmer may be given.

(b) Calculate (i) the number of ways the 3 prizes can be awarded among the 6 helpers, and (ii) the value of each prize.

(c) Determine the official opening time of the festival.

Item

The farmer is setting up a new solar pump to solve the problem of water on his farm. The pump uses a controller switch defined by a number

$$Z = \frac{(2+i)(3-2i)^2}{(1-2i)}$$

For the safety of the pump the size of this switch should be greater than 12 and its angle when switched on must be acute.

The pump sends water to the tank every hour and someone already using it, has records that in the second and third hours, he receives a total of 300 liters of water, while in the fourth and 5th hours, he receives a total of 1200 liters. At this rate the pump sends water such that each hour, the amount of water keeps increasing and multiplying by the same factor each hour. The farmer requires at least 5000 liters in 7 hours, but he doesn't know whether this pump will solve his problem. The pump sends water to the tanks using pipes of length:

$D_1 = \sqrt{(2x + 1)}$ *metres and* $D_2 = \sqrt{(x - 3)}$ *metres*

where a is the cross-section area of the pipes.

The total length of these pipes is 4 metres but the required pressure on the farm can be realised only if the cross-section area is between 3 and 6 inclusive.

Task:

Help the farmer to make valid decisions using mathematical skills

ITEM: SEVEN

In a certain town, trailers and other heavy vechicles move following a particular road pattern described by the curve $3x^2 - 5x + 1 = 0$, whose roots are α and β. However, when the Town clerk of the Town visited the area for inspection, he realized that there was congestion and decided to construct a new road of the same nature as the existing one, with roots being $\alpha + \frac{3}{\beta}$ and $\beta + \frac{3}{\alpha}$

The revenue department in the clerk's office advised the clerk that to construct this new road easily,

After analyzing the length of this new road, the engineering department team found it to be:

$$\frac{2\sqrt{2}}{3-\sqrt{2}} \div \frac{4+\sqrt{2}}{3+\sqrt{2}}\ km$$

However, the Clerk tasked them to express it as a linear expression of the form $a + b\sqrt{2}$.

The Clerk has decided to form a committee of four members among whom is the Mayor of the Town council. He is choosing from a group of four engineers and 5 revenue specialists. The committee can have one or two engineers.

To protect the account where the revenue money has to be placed, he has decided to hide the last value of the password in the soluteion of the equation $2\sqrt{(x+1)} - 3\sqrt{(2x-5)} = \sqrt{x-2}$. But he is not sure of this value and wants this information summarized in report form.

Task:

Help the Clerk to formulate the report for proper documentation.

ITEM: EIGHT

In a rural village, the village development committee agreed to collect funds every month from households to install a community solar power grid. They decided to contribute UGX 20,000 in the first month, and the monthly contribution then increases by UGX 5,000 each month thereafter for 20 months. The committee chair wants to determine whether the planned contributions will be sufficient to meet the budget target for the solar equipment of UGX 50 million, given a total of 400 households participating in the scheme.

The local cooperative has also introduced a productivity bonus for the solar technicians based on their hours worked per week. The total number of hours (x) accumulated by one technician in a week must satisfy the condition: "the square of the hours plus 4,000 must be less than or equal to 125 times the number of hours" for the technician to qualify for the performance bonus. A senior technician has worked a total of 55 hours in a certain week and is unsure whether he qualifies for the bonus.

The maintenance crew consists of 6 junior technicians and 5 senior technicians. The supervisor wants to create a daily maintenance team of 5 technicians with an equal number of junior and senior technicians (or as close as possible to maintain balance). This is intended to ensure proper mentorship and will only be effective if the total number of possible distinct teams exceeds 100.

To improve the safety of the remote monitoring software, the IT department is installing a new encryption system. The strength of the system depends on a security key represented by x units, where for a strong security key, x must be at least 2 units. During testing, the software engineer finds that the overall security rating is obtained by combining two separate measurements: the first measurement is $\log_x 9$; the second measurement is $2 \log_3 x$. The system will function correctly only when the total security rating is equal to 3 units.

Task:

a) Perform a financial feasibility study for the village development committee to determine if they will meet the set budget target.

b) Advise the maintenance supervisor on the effectiveness of the team selection process based on the number of teams possible.

c) Determine the eligibility of the senior technician for the performance bonus.

d) Evaluate the technical viability of the new encryption system.

ITEM: NINE

A fertiliser company supplies a mixture to tea growers.

The total yield over four consecutive seasons is modelled as Total yield $(kg) = (Ax + By)^3$, where A and B are the ratios of fertiliser (x) and water (y) in each container. Each container holds 4 kg of fertiliser and 2 litres of water. The first-season yield was 2744 kg and the fourth-season yield was 64 kg.

The percentage effectiveness, P, of the fertiliser after t months is modelled by $log\ P = log\ P_0 - C\ log(t - 3)$, where C is a concentration constant. A new batch, manufactured on 1 November 2026, has an initial effectiveness of 90% and C = 0.25. The batch is fit for use as long as its effectiveness does not fall below 45%.

The company also wants to site three sales outlets whose positions satisfy $Z^3 - 19Z + 30 = 0$, where Z is a coordinate on the site map.

Task:

(a) Determine which season gave the highest yield.

(b) Recommend, with calculations, the expiry date to print on the new batch.

(c) Find the coordinates of the three sales outlets.

THEME THREE: CALCLUS (SECTION B PART II)

ITEM: TEN

Moses deals in making concrete bricks locally and his brick making process has three stages: excavation of stone dust, making the bricks, and drying the bricks. He has been given a contract to make thousands of bricks.

He has hired an excavator machine and his target is to excavate at least 4500 m³ of stone dust. This machine can excavate stone dust such that the volume excavated after time t days of continuous excavation is modeled by:

$v^2 + t^2 - 6t - 16 = 0$

where v is in thousands of cubic metres.

His brick molding machine can mold the bricks at a rate of:

$\frac{dN}{dt} = \frac{(150t+500)}{(t-1)(t+5)}$

where N is the number of bricks in hundreds and t is time in days and he has only 20 days to mold at least 30,000 bricks to be presented as samples.

These bricks cannot be dried under direct sunlight; therefore, he has 12 days to dry them. Each brick is manufactured at a temperature of 10°C and dries to a temperature of 20°C in 4 days in a shade whose temperature is 40°C. The bricks are drying at a rate proportional to the excess temperature of the shade over that of the bricks. The bricks can be loaded for transportation if they dry to a temperature not less than 29°C.

Task:

Using mathematical calculations help Moses to make valid decisions.

ITEM: ELEVEN

A company dealing in ceramics wants to design a new vessel whose edge is defined by the curve $y = 8 - x^3$. The vessel surface is to be formed by rotating the area cut off by this edge and the axes about the x-axis and then made hollow on the inside. The company wants to determine the area enclosed by the edge of the vessel and the axes. The company wishes that this area does not exceed 13 sq. units, or else the available machinery will need adjustments. The company wishes that the volume of the vessel should not be less than 250 cubic units to meet the available capacity standards.

The company has projected that the rate of increase in its earnings will be directly proportional to the product of the square of its earnings increased by twenty times the earnings, and twice the time less 1. Time is determined from the start of manufacturing of this new vessel. The company experts have suggested that at the start of manufacturing of this vessel, the total earnings will be 50 millions and after half a year, the total earnings will have doubled. The company wants to know what the earnings will be after two years of manufacturing this vessel. The company wishes that if these earnings after two years do not exceed 150 millions, then it will have to venture into another product.

Task:

Help the company obtain the right information to direct its decisions.

ITEM: TWELVE

A pineapple farmer has been facing a problem of marketing his pineapples and decided to start a small factory of making and packaging pineapple juice. He wishes to pack his juice in a cylindrical tin whose height is twice it's radius. However, the tins on market have an error of 1% in their radius and in the juice market the acceptable percentage error in volume is at most 4%. He doesn't know whether to use the tin or not.

He also bought a machine to be used for filling the juice into a 310ml tin and the machines operate at a rate of

$\frac{dV}{dt} = t^2 - 2t - 40 \quad for\ 0 \leq t \leq 15$

where v is the amount of juice in cm^3 filled in time t seconds.

The machine passes the test if it can fill at most 97% of the volume of the tin so as to leave some space for the preservative gas.

He also plans to buy a preservative which can keep the juice fresh for at least 10 days. The preservative on market decomposes in the drink at a rate proportional to its concentration if applied in the drink and its effect stops when its concentration goes below 40%. If the preservative is added to the drink, its concentration drops to 90% in one day.

Tasks: Using mathematical calculations;

a) Help the farmer to know whether the tin will be of acceptable volume.
b) Determine whether the machine he needs to buy will pass the test.
c) Determine whether the preservative will help to keep the drink fresh for the required time.

ITEM: THIRTEEN

Suzan has discovered that there are many nursery schools in her locality and has decided to start a small business of making ice cream. She purchased molds that can make conical ice cream packs; however, she discovered that the vertical angle θ is 32° instead of 30°. This will result in losses if the corresponding error in the radius of the cone exceeds 0.2 cm.

The relationship between the radius of the cone and the vertical angle is: $r = 9 \tan\theta$.

She plans to use two primary ingredients, milk and sugar, mixed according to the formula; $ds/dm = m \ln m$, where m is amount of milk in litres and s is amount of sugar in grams, with 1 litre of milk requiring 30 g of sugar. She will only use this formula if her expected daily usage of 50 litres of milk requires less than 5 kg of sugar.

Suzan plans to use a mixture of chocolate and vanilla flavors. Experts informed her that this mixture degrades at a rate proportional to the amount remaining. Specifically, 5% of the flavor mixture degrades in 3 hours. She will only consider this mixture if, after 12 hours, not more than 20% of the flavor mixture has degraded.

Task

Help Suzan to make valid decisions on her new ice cream business plans using your calculus skills.

ITEM: FOURTEEN

A landowner wants to fence a rectangular plot of area 32 000 m² using wire mesh. One side of the plot lies along an existing wall, so only the other three sides need fencing, with a 10 m gate left on the side opposite the wall.

The rate at which the fence is installed after t hours is given by $\frac{dL}{dt} = 9 - 0.05t$, for $0 \leq t \leq 10$, where L is the length (metres) fenced.

The plot has a depression requiring 150 m³ of soil to fill it, and an ant-hill at its centre whose cross-section is modelled by $y = 20 - 4x^2$, where x is horizontal distance from the centre (m) and y is height above the field (m). The soil removed in levelling the ant-hill is to be used to fill the depression.

Task:

(a)(i) Calculate the minimum length of wire mesh needed to fence the plot.

(ii) Find the number of days required to install it.

(b)(i) Sketch the cross-section of the ant-hill.

(ii) Determine whether the excavated soil will completely fill the depression.

ITEM: FIFTEEN

Senior Six students are preparing exhibits: the girls mould items from clay, the boys compete on packing speed.

Fatuma made a conical mould of radius 8 cm and height 90 cm and wants to reshape it into a cylinder of the largest possible volume.

Grace intended a spherical mould of radius 12 cm but produced one of radius 11.7 cm. Only moulds whose volume error does not exceed 6% will be approved.

Kato is practising packing 150 bottles into a box at a rate proportional to the square root of the bottles remaining. In his last practice he packed 24 bottles in the first minute; last year's winner finished in 15 minutes.

Task:

(a) Find the volume of the largest cylinder Fatuma can mould from her cone.

(b) Determine whether Grace's mould will be approved.

(c) Assess whether Kato is likely to beat last year's winning time.

P425/2
PRINCIPAL MATHEMATICS
Paper 2 2026

3 hours

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Respond to **three** *items* **in all**.

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Begin each item on a fresh page.

All *necessary working* **must** *be shown clearly.*

Graph paper is provided.
Where necessary, use $g = 9.8ms^{-2}$.

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

P425/2 PRINCIPAL MATHEMATICS PAPER TWO ASSESSMENT FORMAT					
SECTION	PARTS	THEME/ CONSTRUCT	TOPICS	NUMBER OF ITEMS SET	ITEMS ATTEMPTED
A	COMPULSORY	NUMERICAL METHODS ITEM 1	1.Iterative methods 2.Error analysis 3.Trapezium Rule 4.Flow charts	1	1
B	I	DATA ANALYSIS AND PROBABILITY ITEM 2 AND ITEM 3	1.Descriptive Statistics 2.Correlation and scatter diagram 3.Probability theory 4.Probability distribution 5.Random Variables 6.Sampling Distribution	2	1
	II	MECHANICS ITEM 4 AND ITEM 5	1.Resultant and components of forces 2.Friction 3.Connected particles 4.Resultant velocity 5.Relative motion 6.Projectile	2	1
TOTAL ITEMS				5	3

THEME ONE: NUMERICAL METHODS (SECTION A)

ITEM 1

A district local government is constructing a community water supply system to provide clean water to a newly established health centre. Before approving the construction, the District Engineer must verify the accuracy of the proposed water storage tank

dimensions, assess the adequacy of the drainage channel, and confirm the required diameter of the main supply pipe.

The tank has a radius of 2.5 m, correct to one decimal place. The engineer estimates that the maximum relative absolute error in the calculated volume of the tank is 0.125. Before approving construction, the engineer wishes to determine whether the measured height of the tank is sufficiently accurate. If the maximum relative absolute error in the measured height exceeds 0.10, the height of the tank will be measured again before construction begins. Otherwise, the existing measurements will be approved for construction.

To protect the water tank from flooding during the rainy season, the District Engineer designed a drainage channel. The cross-sectional profile of the channel is modelled by

$$y = \frac{50\text{x} + 27\ tan^{-1}\ \text{x}}{1+x^2}$$

where x and y are measured in metres.

The engineer recorded the vertical distances (ordinates) from the ground level to the channel profile at five equally spaced positions between $x = 0\ and\ x = 6$, inclusive, and used these measurements to estimate the cross-sectional area correct to three decimal places. If the estimated cross-sectional area exceeds 120 m^2, the drainage channel cross-section will be enlarged to improve storm-water discharge. Otherwise, the current drainage design will be maintained.

To ensure adequate water pressure throughout the distribution network, the internal diameter (in centimetres) of the main supply pipe must satisfy the equation:

$$e^x + x - 4 = 0$$

Preliminary calculations indicate that the required pipe diameter lies between 1.0 cm and 1.2 cm. Before purchasing the pipes, the engineer wishes to obtain a better approximation to the required diameter correct to three significant figures. If the better approximation is less than 1.1 cm, a standard PVC pipe will be used. Otherwise, a high-pressure commercial pipe will be purchased

Task:

Using appropriate mathematical methods, advise the District Engineer on the suitability of the proposed designs for the community water supply project before construction begin.

ITEM 2

A landlord is upgrading the compound of a newly completed rental estate to improve drainage and security. The project involves paving the compound and installing a surveillance camera at a strategic location.

The paving contractor supplies rectangular concrete pavers whose advertised top surface dimensions are 10.45 cm by 3.6 cm, each measured to the nearest decimal place. Before approving the delivery, the site engineer must verify whether the pavers meet the required quality standards. According to the project specifications, the consignment should be rejected if the percentage error in the area of a paver exceeds 1.5%.

To estimate the section of the compound to be paved, the engineer models the boundary of the paved region using the curves

$y = x \sin x \quad and \quad y = e^{-2x}, \quad for\ 0 \leq x \leq 2,$

where both x and y are measured in tens of metres.

For security purposes, the landlord intends to install a surveillance camera at the point where the two boundary curves intersect along the x-axis. Since an exact algebraic solution is difficult to obtain, the engineer will determine the point of the intersection graphically correct to 1 decimal place.

The landlord has allocated a budget sufficient to pave not more than 25 m² of the compound. To determine whether the planned paved section falls within the available budget, the engineer must estimate the enclosed area between the two curves using a suitable method with four equal subdivisions correct to 3 decimal places.

Task.

As a mathematics student help the land lord make informed decisions.

ITEM 3

National Water and Sewerage Cooperation (NWSC) is encountered with a problem of the volume capacity of their cylindrical pipes that supply water to their customers. They consulted their engineer who found out that the diameter is 0.62metres and length 6.2metres. There was need to know the range between which the volume of cylindrical pipe lie, the engineer advised the NWSC manager that for the cylindrical pipes to provide sufficient water supply to its customers, the range should be between 1.6 and 2.0.

The NWSC was also encountered with a problem of location of a point at which to locate a pumping machine along a linear bank of a river that stretches from 3km to 5km. Previously data collected by the engineering department had indicated that this area was bounded by a line ($y = 4x + 6$) and another curve ($y = 3e^{0.5x}$). The surveying department

noticed that using a graphical method the point of intersection between the above two curves that lie along the line from 3km to 5km above would be a suitable approximate point for the location of the pumping machine and required that it should be rounded off to 1 decimal place, use a suitable iterative method to get the exact value, if this point lies between 3.5 and 4, it would be a suitable approximate point at which to install the pumping machine.

The NWSC was also encountered with a problem of establishing the costs and distances involved in transportation of their water installation materials. It had been noted for distances of 30km, 40km, and 50km the costs were sh. 60,000/=, sh. 100,000/= and sh. 130,000/= respectively. They were approached by two of their customers, one paid sh. 150,000/= and another one was to be left at a distance of 46km. There was need to use a suitable linear method to estimate for them the distance and the cost. The second customer complained that he had been overcharged by paying sh. 120,000/= and there was need to justify his claim.

TASK:

As a mathematics student, analyze the problems of NWSC by giving an interpretation, methodology and justification.

ITEM 4

The Uganda National Roads Authority (UNRA) is upgrading a 50km road that connects a trading centre to a lake community. A Sino truck transports gravel from a quarry and off loads it at different points along the road according to the condition of the road surface.

The following offloading records were made along the route. At the trading centre 4 tonnes were offloaded. At the swampy area 40km from the community 3.4 tonnes were offloaded. At the beginning of the hill 20km from the centre 2.8 tonnes were offloaded. 30km from the centre 2.2 tonnes were offloaded. At a distance of 10km from the lake community 1.6 tonnes were offloaded and at the lake community 1.0 tonnes were offloaded. The roads supervisor wants to plan the distribution of stones more effectively for the remaining sections of the road. The engineer wanted to offload stones at a distance of 9km from the community and also wanted to offload 2.09 tonnes of stones

Engineers also established the relationship between the quantity of stones offloaded, y tonnes, and the distance x in Kilometers as $e^x cosx + 1 = 0$ for the section $1 \leq x \leq 2$ (kilometers) for the road design in that particular area. The engineer wanted to perform a dry run to identify actual location to a degree of accuracy of two decimal places.

Task

(a) Advise the supervisor on the quantity of stones to be offloaded and the location for offloading from the centre.

(b) (i) Help the supervisor to establish the road design graphically, and distance where offload was made. ii) Advise the engineer on the most accurate location.

ITEM 5

A rural growth centre is short of piped water, and the Sub-county Council has agreed to draw extra water from a nearby channel that is exactly 24 m wide. The Council has instructed its Water Engineer to install a pump that will lift water from the channel to the centre's storage tank.

To plan for this, the engineer has modelled the channel's depth $D(x)$ (in metres) by the function

$$D(x) = 14.4 - 0.10(x - 12)^2, \text{ for } 0 \le x \le 24,$$

where x is the distance, in metres, from one bank of the channel.

For the pump to draw water efficiently, the cross-sectional area of the channel must be at least 210 m², and the engineer is confident that his model meets this requirement.

He explains that if he takes readings at regular 6 m intervals from the bank and applies an approximate integration method, he can readily show that the model satisfies the minimum cross-sectional area needed. He adds that the same conclusion can be reached by evaluating the exact area under the curve.

The base of the pump, however, needs to sit where the depth D(x) is at least 9 m. Starting with an initial estimate of x = 5 m from the bank, the engineer claims he can pin down the required distance so that the tolerance in his estimate does not exceed 0.08 m, ensuring the pump base is stable.

Before the installation goes ahead, the Sub-county Council wants to be satisfied — using both an approximate and an exact method — that the channel truly offers the minimum cross-sectional area needed, and it wants to know precisely how far from the bank the pump should be positioned.

Task:

(a) Give the Council the verification it needs.

(b) Determine, and report to the Council, the exact distance from the bank at which the pump should be installed.

THEME TWO: DATA ANALYSIS AND PROBABILITY (SECTION B Part I)

ITEM 6

The sports department in a certain Radio station is preparing for the final match of a popular national football tournament involving Team A and Team B. The department is conducting a study on spectators' expectations and match outcomes to assist tournament organizers in planning security, seating arrangements, refreshments, and media coverage.

A survey was conducted among football fans, and the results are summarized below.

Predicted number of goals in the final match	**0**	**1**	**2**	**3**	**4**	**5**	**6**
Number of fans (in hundreds)	5	15	25	36	26	14	4

The tournament organizers would like to determine the median, mean, and standard deviation of the predicted number of goals. The organizers have agreed that if the median predicted number of goals is more than four, additional entertainment activities should be arranged during halftime because spectators are likely to be excited.

The sports analysts have also estimated that the probability of Team A scoring a goal during the match is 0.60, while the probability of Team B scoring a goal is 0.45. The organizers wish to determine the expected number of teams scoring. If the expected number of teams scoring exceeds 1.5, additional broadcasting and media resources will be allocated to cover what is anticipated to be an exciting match.

Furthermore, historical performance data indicate that the probability of Team A winning the final match is 0.73. At one regional viewing centre, 52 football fans have participated in a prediction challenge by forecasting that Team A will win the match. The organizers would like to determine the probability that at least 40 of these fans predict correctly. If this probability exceeds 0.30, the organizers will prepare recognition awards for participants who made accurate predictions.

Task.

As a sports data analyst, use the information provided to help the tournament organizers make informed decisions.

ITEM 7

A secondary school administration carried out an evaluation of students' performance in Economics to determine whether there had been any improvement in academic achievement over the course of the term. To do this, the administration compared the

Beginning of Term (BOT) examination scores to students with their corresponding End of Term (EOT) examination scores. The results are shown in the table below:

BOT(X)	70	60	58	45	80	89	45	72	58	51
EOT(Y)	65	53	55	47	75	86	58	70	65	44

The administration wishes to illustrate the relationship between the BOT and EOT scores using an appropriate graph and determine whether a statistically significant relationship exists between the two sets of scores.

If the relationship is found to be significant at the 5% level of significance, the administration intends to organize an academic review meeting with the Economics teachers to discuss students' performance.

In addition, the Head of the Economics Department has been asked to prepare a statistical summary of the End of Term examination results for the entire class. The grouped frequency distribution of the EOT marks is presented below:

Marks	35– 45	−50	−< 60	−< 65	−< 75	−< 80	−< 95
Number of students	4	13	25	30	20	15	3

The administration would like to determine the modal mark for the End of Term examination. If the modal mark is less than 55, a meeting will be arranged with the students to discuss strategies for improving their academic performance.

Furthermore, the Head of the Economics Department has advised that the End of Term examination scores are approximately normally distributed. Based on this assumption, the administration wishes to identify the group of students who scored more than 85 marks, as these students will be appointed as leaders of the various Economics discussion groups.

Task:

Help the administration obtain the information needed to make informed decisions.

ITEM 8

A District Education Officer (DEO) is reviewing how well science laboratories are being used in his district. During a visit to a school offering both O-Level and A-Level classes, he learns that the probability a given student made use of the Physics, Chemistry and Biology laboratories is 0.35, 0.45 and 0.20 respectively.

He further learns that the probability a student is in A-Level, given that the student used the Physics, Chemistry or Biology laboratory, is 0.60, 0.80 and 0.55 respectively.

The DEO intends to recommend to the district council that new laboratories be built if the proportion of students who used the Chemistry laboratory, given that they are A-Level students, exceeds 0.45, and if the expected number of laboratories used per student is at least 1.8.

In addition, the DEO has sampled 12 students from the school and wishes to work out the probability that at least half of them used the Chemistry laboratory. Should this probability exceed 0.65, additional laboratory equipment will be supplied.

Task:

Help the DEO to:

(a) reach a well-supported recommendation on whether new laboratories should be built;

(b) decide whether the extra equipment should be supplied.

ITEM 9

In Uganda, the cases of road accidents are increasing because of reckless crossing and over speeding of pedestrians and drivers respectively. Because of this, traffic lights have been mounted on most busy roads. A case in point is the Kampala northern bypass whereby at a given point the speeds of the cars are normally distributed. Police found out that 95% cars travelled at a speed less than 85 kmh^{-1} and 10% travelled at a speed less than 55 kmh^{-1}.

On a certain day, an operation by police was carried out on 5 cars to find out how often these accidents occur. They came to realize that the chance of an accident to occur at this point was twice that of an accident not to occur. The O.C traffic instructed that if the chance that at least 3 accidents occurring in a week exceed 0.49, new road humps will be constructed unless otherwise.

In the process, many cars were impounded and each car had its issues depending on the mechanical conditions found on it. These conditions followed a continuous random variable that was uniformly distributed in the interval $a\ to\ b$. The mechanical conditions for car A had a lower quartile of 5 and an upper quartile of 9. It is a policy by the police to imprison a driver if the mechanical issues of the car lie in the interval between 2 and 12.

Tasks

As a senior six student if consulted, come up with a clear report on the following backed by calculations:

(a) The average speed and proportion of cars that exceeded 50 kmh^{-1}.

(b) Whether new road humps will be constructed or not.

(c) Whether it was possible for the driver of car A to be imprisoned or not.

ITEM 10

A district education office organized a Career Guidance and Skills Development Workshop at a regional training centre. The workshop featured career talks, mentorship sessions, practical demonstrations, and counselling services. To effectively plan refreshments and participant welfare services, the organizing committee investigated the amount of time participants spent at the workshop.

The time, *x* (in hours), that a participant spends at the workshop is modelled by the probability density function

$$f(x) = \begin{cases} ax, & 0 < x < 3 \\ 3a(4-x), & 3 < x < 4 \\ 0, & otherwise \end{cases}$$

To promote participant welfare, the organizing committee established the following guidelines:

- Participants who remain at the workshop for longer than the expected (mean) time will receive two rounds of refreshments.
- Participants who stay for more than 3.5 hours will receive two rounds of refreshments and a takeaway snack package.

The committee wishes to determine the percentage of participants who qualify for:

- Two rounds of refreshments only.
- Two rounds of refreshments and a takeaway snack package.

During the workshop, the access road leading to the training centre became heavily congested. To ensure the safety of participants, local traffic officers monitored the speeds of vehicles entering the venue.

The vehicle speeds are assumed to be normally distributed, and the following information is available:

- 5% of the vehicles travel at speeds greater than 85 m/s.

- 10% of the vehicles travel at speeds less than 25 m/s.

For the safety of pedestrians and workshop attendees, a speed limit of 60 m/s was imposed along the access road. The authorities agreed that if more than 25% of the vehicles exceed this speed limit, additional traffic-control measures, such as speed humps and traffic wardens, would be introduced.

A random sample of 100 vehicles was observed, and the organizing committee wishes to determine the 95% confidence interval around the mean speed.

Tasks

As a mathematics student, use the information provided to assist the;

(a) Organizing committee plan for rounds of refreshment and a takeaway snack

(b) Authorities to determine whether to add traffic control measures

(c) Organizing committee determine the required confidence interval.

ITEM 11

A teachers' association in a certain district carried out a mock chemistry examination. To evaluate how learners performed, the association analysed a sample of 90 candidates.

The scores obtained are summarised in the table below.

Scores (%)	**No. of students**
0 - < 20	12
20 - < 40	25
40 - < 50	28
50 - < 60	18
60 - < 90	7

According to the association's benchmark, performance is judged good when both the mean score and the modal score are at least 40%, and the examination is judged reliable when the standard deviation of the scores does not exceed 12%.

The association wants to establish whether performance was good, and how reliable the examination was.

Across the whole district, 600 candidates sat the same chemistry examination, and their scores were approximately normally distributed.

The district chemistry inspector has been asked to analyse the district-wide results and, in particular, to determine:

(i) the cut-off marks for a distinction and for a fail, given that the top 12% of candidates are awarded a distinction and the bottom 18% fail;

(ii) the number of candidates who passed the examination and the number who obtained a distinction;

(iii) the interval within which the mean score of all district candidates lies, at a 98% level of confidence.

Task:

(a) Using your calculated results, report on:

(i) how well the candidates performed in chemistry;

(ii) how reliable the examination was.

(b) Respond to the district chemistry inspector's three requirements above.

THEME THREE: MECHANICS (SECTION B Part II)

ITEM 12

A transport company has been contracted to deliver essential supplies from Masaka to a warehouse on an island in Lake Victoria. The delivery must be completed within a strict schedule because the supplies are required for a community development project.

Early in the morning, a truck driver departs from a distribution centre in Masaka carrying the supplies. The driver has been informed that the ferry connecting the mainland to the island is scheduled to depart from the port in 10 minutes' time. Missing the ferry would delay the delivery by several hours and increase transportation costs. The truck is initially 20 km from the ferry port. It has a mass of 3.5 tonnes and moves along a rough horizontal road. The engine exerts a constant driving force of 1.5 kN, while resistive forces due to friction and air resistance amount to 0.25 N per kilogram of the truck's mass. The transport manager wishes to determine whether the truck can reach the port before the ferry departs.

Upon arrival at the port, the truck is loaded onto the ferry for transportation to the island. The ferry captain must navigate directly to an island located due west of the port. However, the captain knows that the lake has a current flowing at 3 km h^{-1} in the direction S28°E. The ferry is capable of moving at 10 km h^{-1} in still water. To ensure that the ferry reaches the island without drifting off course, the captain must determine the direction in which the ferry should be steered.

After reaching the island, the truck is driven to a warehouse where the supplies are unloaded. Some packages are discharged from the back of the truck by allowing them to leave the loading platform horizontally at a speed of 10 m s^{-1} from a height of 2 m above

the ground. To prevent damage to the supplies, safety guidelines require that the packages should not strike the ground with a speed exceeding 15 m/s. The warehouse supervisor therefore wishes to assess whether the unloading procedure satisfies the safety requirements.

Tasks

As a mathematic student help the;

(a) transport manager to determine whether the truck can reach the port before the ferry departs.

(b) captain determine the direction in which the ferry should be steered.

(c) warehouse supervisor to assess whether the unloading is safe

ITEM 13

Peter is a very concerned parent. One day, he bought two toy cars , one of mass 1 kg and the other of mass 4 kg. To his surprise, he found his 5-year-old boy trying to combine the two toys, but the strings kept breaking and sometimes the heavier toy would fall on his feet. This always made the little boy cry a lot while playing. Peter decided to get him a rough plane and tried to arrange the setup as follows:

For the safety of the child, he placed the 1 kg mass toy on a 4 m long rough plane which has a coefficient of friction 0.25. A smooth fixed pulley was placed at the edge of the plane. A light inextensible string was connected to the 1 kg mass and allowed to pass over the pulley to a 4 kg mass which hangs freely at 2.5 m above the horizontal ground. Since he wanted the boy to enjoy the movement of the toys, he tilted the plane such that the angle of inclination θ with the horizontal is $\sin^{-1}\frac{3}{5}$.

The system is released from rest. In order for Peter's son not to cry again, the following conditions must be met:

Condition 1: Acceleration of the system must be less than 7 $m\ s^{-2}$

Condition 2: The tension in the string must not exceed 15 N

Condition 3: The maximum kinetic energy with which the 4 kg mass hits the ground must be less than 70 J

Task

As a mathematics student, determine whether the setup Peter made will stop his son from crying and justify your responses.

ITEM 14

The Ministry of Works and Transport is investigating the safety of a steep section of a certain road during dry and rainy seasons for heavy trucks. The section of road is inclined at an angle of 12° to the horizontal, and the estimated road section is determined at an angle of 12° to the horizontal.

Coefficients of static friction between the tyres of heavy trucks and the road surface are as shown below:

Road condition	Coefficient of friction
Dry season	0.30
Rainy season	0.12

As part of the investigation, a fully loaded supply truck of mass 15 tonnes is positioned on the slope and is initially at rest with brakes engaged and engine switched off. The truck's parking brake is then released. If the truck remains stationary after the parking brake is released, the road will remain open to heavy vehicles during that season. Otherwise, a second safety test will be carried out to determine whether disabled heavy vehicles can be recovered safely.

For the second test, a breakdown truck of mass 20 tonnes, having a maximum engine driving force of 120 kN, is connected to the supply truck using a tow cable with a maximum safe working tension of 50 kN. If the breakdown truck is able to pull the supply truck up the slope without the tension in the tow cable exceeding its maximum safe working tension, the road will remain open to heavy vehicles during that season, provided that a fully equipped breakdown truck is kept on standby at the site. Otherwise, the road will be temporarily closed to heavy vehicles during that season until a higher-capacity cable or more effective recovery arrangements are put in place.

Hint:

Assume that:

- Both trucks are on the same 12° incline.
- The tow cable is light, inextensible, and remains parallel to the road surface.
- The coefficient of friction between the tyres and the road remains constant for each road condition throughout the investigation.
- Rolling resistance and air resistance are negligible. Acceleration due to gravity, g = 9.8 m s^{-2}.

Task

Using appropriate principles of mechanics and mathematical justifications, advise the Ministry of Works and Transport on the appropriate measures for the safe operation and recovery of heavy vehicles along this section during the dry and rainy seasons.

ITEM 15

A businessman wants to install a regular pentagon solar panel structure. He consults an engineer who finds out that the regular pentagon solar panel structure has a mass of 10 kg and a length of 2 m, with forces of 6 N, 13 N, 5 N, 2 N, and 3 N acting along the sides AB, BC, CD, DE, $and\ AE$ respectively. The directions of the forces are in the order of the letters. The engineer took AB as the reference horizontal axis. The engineer advised that there was need to know the resultant force and that it should not exceed 18 N for the cost of the solar panel ($ABCDE$) to be affordable. It was also important to know the direction of the resultant force.

The engineer informed the businessman that the regular pentagon solar panel was installed at the top of the roof at a distance 4 m from the edge of the roof that is inclined at 30° to the horizontal. There was need to know the velocity with which it reaches the edge in case it is detached from the roof and slides down a rough surface with a coefficient of friction of $\frac{1}{4}$. He advised that the velocity should not go beyond 10 $m\ s^{-1}$, otherwise there would be loss of energy.

The engineer noted that the regular pentagon solar panel was projected below the horizontal at an angle of 30° below the horizontal with a velocity above. The height above the ground to the edge was measured as 4 m by the engineer. There was need to know the horizontal distance from the height to where the rectangular solar panel hits the ground. He preferred that it does not go beyond 3 m since there are usually children playing nearby. There is need to know the safety of these children.

Task

As a mathematics student, analyze and justify the regular pentagon solar panel problems to the businessman.

ITEM 16

A large logistics company operates a fleet of cargo vehicles transporting goods between major distribution centres in Uganda. To improve efficiency and ensure the safety of goods in transit, the company has installed a real-time vector-based tracking system that monitors the positions and velocities of vehicles moving in three-dimensional space.

At 12 noon on a particular day:

- A cargo trailer departs from the company's central warehouse from position (i + 7j) km, moving with constant velocity (6i + 2j) km h^{-1}
- A delivery truck departs from a regional distribution centre from position (6i + 4j) km with constant velocity (−4i + 8j) km h^{-1}

To prevent delays and possible loss of goods, the logistics control centre must determine whether the two vehicles will occupy the same position at the same time. If this occurs, the control centre must identify the exact location so that alternative routing arrangements can be made.

Later that week, one of the company's loaded trailers developed a mechanical fault while transporting goods to a distribution centre. A recovery vehicle was dispatched to tow the trailer to the warehouse.

The trailer has a mass of 8 tonnes, while the recovery vehicle has a mass of 2.5 tonnes. The recovery vehicle exerts a driving force of 4 kN while moving on a horizontal road. The coefficient of friction between the vehicles and the road surface is 0.2.

To ensure the safety of personnel and cargo, the logistics manager wishes to determine the tension in the towing system. The recovery operation can be carried out efficiently if the tension is more than 3 kN.

Tasks

(a) Use information above to determine whether the two vehicles occupy the same position at the same time and find the exact position and time at which this occurs.

(b) Justify the effectiveness of the recovery operation.

ITEM 17

At exactly 11:00 am, an emergency response coordinator was monitoring two rescue units during a disaster response exercise. The emergency control centre was taken as the origin.

A taxi was carrying medical supplies while a rescue truck was transporting emergency equipment. Their initial positions and constant velocities at 11:00 am were recorded as follows:

Unit	Position vector	Velocity
Taxi	$(5i + 2j)\ km$	$(15i + 10j)\ km/h$
Rescue truck	$(7i + 7j)\ km$	$(9i - 5j)\ km/h$

The coordinator wanted to determine whether the two rescue units would arrive at the same location if they continued moving with their recorded velocities. If they would, determine the time when they would meet and the total distance travelled by the rescue truck by that time.

Later, the taxi developed a mechanical fault and had to be transported to the maintenance centre. A recovery truck of mass 1800 kg was used to tow the damaged taxi, which was of mass 1200 kg, using a tow bar along a rough horizontal road.

The recovery truck applied a driving force of 6000 N. The damaged taxi experienced friction with a coefficient of 0.2, while the recovery truck experienced a resistive force of 800 N.

For the towing operation to be considered safe, the acceleration must be less than 2 m s^{-2} and the tension in the rope must not exceed 3500 N. Otherwise, a heavier recovery vehicle would be required.

Task

As a student of mathematics, help the coordinator to make informed decisions.

ITEM 18

A game ranger at Murchison Falls National Park spots a buffalo running along a straight line towards River Nile. The ranger observes it at post A. At the moment of observation (t = 0), the buffalo is at point B,120 m east of post A and is running due north at a constant speed of 20 m s^{-1} relative to the ground.

Simultaneously, a rescue drone is launched from post A to move as close as possible to the buffalo's path. The drone flies at a constant speed of 10 m s^{-1} relative to the ground, maintaining a fixed direction.

At the instant when the drone is $60\sqrt{3}\ m$ horizontally away from the buffalo, it projects a tranquilizer dart horizontally towards the buffalo. The dart is released from a height 80 m above the ground with an initial speed $u\ m\ s^{-1}$. To avoid injuring the buffalo, the speed with which the dart hits the buffalo should not exceed 47 m s^{-1}.

When the buffalo was hit, it got shocked and jumped upwards a distance of 0.45 m high. The game ranger wishes to get analysis of the above information mathematically for better reporting.

Tasks

As a mathematics student, come up with necessary calculations to help the ranger establish:

(a) The closest distance reached by the drone and the time it took to be achieved.

(b) If the buffalo was injured or not.

(c) The initial velocity with which the buffalo jumped.

ITEM 19

A construction firm is renovating an office block in town. To hoist heavy crates up to a service platform, the firm's technical crew has rigged a pulley system made up of a light, inextensible cable fixed to a roof beam.

The cable passes under a smooth movable pulley carrying a counterweight of 60 kg, then over a smooth fixed pulley mounted on the service platform. A crate of mass 25 kg, which is to be raised to the platform, is attached to the other end of the cable. All sections of the cable stay in the vertical plane while the system operates.

The system is currently held at rest, and the crew is about to let it run.

For the safety of pedestrians, a safe ground zone of 6 m from the service platform has been marked off in case any debris falls.

Before the system is put into operation, the engineering team requires that it satisfy the following:

(i) for stability, the acceleration of the system must not exceed 2.5 ms^{-2};

(ii) the cable is rated to withstand a maximum tension of 350 N;

(iii) if released from rest, the crew needs to know how far the movable pulley will have travelled after 6 seconds.

Just as the system is being set up, a tool box slides off the edge of the platform horizontally at 3 ms^{-1}, falling from a height of 25 m above the ground, which alarms the crew about the safety of anyone nearby.

Task:

(a) Determine whether the system meets the requirements set out above, and give a reasoned decision.

(b) Assess how safe pedestrians near the site are.

END

WISHING YOU THE BEST HARVEST